Chapter 27

Coalitional Games with Transferable Utility

In the previous chapter, we have studied the two person bargaining problem where we have explored the effect of cooperation between two players. In this chapter, we introduce multi-player coalitional games. In particular, we discuss *transferable utility (TU) games*, also called characteristic form games. We present several examples of these games to bring out the use of TU games in capturing cooperative interactions among rational, intelligent players.

27.1 Multi-Person Cooperative Games

Let $N = \{1, \ldots, n\}$, as usual, be the set of players. We have already seen the Nash bargaining solution for a two player game. It would be interesting to see what this solution would look like for an n-player game with $n > 2$. Let F be the set of feasible allocations that the players can get if they all work together. Let us assume that F is a closed convex subset of $\mathbb{R}^n$. Suppose $(v_1, \ldots, v_n)$ is the disagreement payoff allocation or default payoff allocation the players would expect to get if they do not cooperate. Also assume that the set $\{(y_1, \ldots, y_n) \in F : y_i \geq v_i \ \forall i \in N\}$ is non-empty and bounded. The pair $(F, (v_1, \ldots, v_n))$ is then called an *n-person bargaining problem*. The bargaining problem $(F, (v_1, \ldots, v_n))$ is said to be *essential* if there exists $y \in F$ such that $y_i > v_i \ \forall i \in N$.

Suppose $(F, (v_1, \ldots, v_n))$ is essential. Then its Nash bargaining solution can be defined to be the unique strongly efficient allocation vector that maximizes the product $(x_1 - v_1) \ldots (x_n - v_n)$ over all vectors $x \in F$ such that $x_i \geq v_i \ \forall i \in N$. However, this Nash bargaining solution may ignore the possibility of cooperation among proper subsets of the players as shown in some of the examples below. Consequently for $n > 2$, the Nash bargaining solution may not give a satisfactory solution. So we have to look for more appropriate solution concepts.

Divide the Dollar Problem

Example 27.1 (Divide the Dollar Game - Version 1). In this problem, $N = \{1, 2, 3\}$. The players wish to divide a total wealth of 300 (to be regarded as a real number) among themselves. Each player can propose a payoff such that no player's payoff is negative

and the sum of all the payoffs does not exceed 300. The strategy sets can therefore be defined as follows:

$$S_1 = S_2 = S_3 = \left\{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 + x_2 + x_3 \leq 300;\ \ x_1 \geq 0;\ x_2 \geq 0,\ x_3 \geq 0\right\}$$

Assume that the players will get a zero payoff unless all three players propose the same allocation. That is, for $i = 1, 2, 3$,

$$\begin{aligned} u_i(s_1, s_2, s_3) &= x_i \text{ if } s_1 = s_2 = s_3 = (x_1, x_2, x_3) \\ &= 0 \text{ otherwise} \end{aligned}$$

Note that the players can achieve any allocation in which their payoffs are non-negative and sum to ≤ 300. The minimum guaranteed wealth is 0 for each player. The above game can therefore be described as a three person bargaining problem (F, v) where:

$$\begin{aligned} F &= \left\{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 + x_2 + x_3 \leq 300, x_1 \geq 0;\ x_2 \geq 0,\ x_3 \geq 0\right\} \\ v &= (v_1, v_2, v_3) = (0, 0, 0) \end{aligned}$$

The Nash bargaining solution for this problem is $x = (100, 100, 100)$, which is clearly a reasonable outcome for this situation. □

Example 27.2 (Divide the Dollar Game - Version 2). This is a variant of Version 1 with the difference that players get a zero payoff unless players 1 and 2 propose the same allocation in which case the allocation proposed by players 1 and 2 is enforced. That is, for $i = 1, 2, 3$,

$$\begin{aligned} u_i(s_1, s_2, s_3) &= x_i \text{ if } s_1 = s_2 = (x_1, x_2, x_3) \\ &= 0 \text{ otherwise} \end{aligned}$$

The same bargaining problem (F, v) as in Version 1 would describe the situation here and hence the Nash bargaining solution for this problem also is $x = (100, 100, 100)$. This solution looks unsatisfactory because players 1 and 2 together determine the payoff allocation and player 3 is not involved in the decision. So, we would expect players 1 and 2 to divide the payoff equally between them, leading to the allocation $(150, 150, 0)$. Another viewpoint which supports this argument is as follows. Suppose 1 and 2 ignore 3 and engage in a two person cooperative game. The resulting two person game would have the Nash bargaining solution that divides 300 equally between 1 and 2.

Having noted the above, there are a few reasons for arguing in favor of the solution $(100, 100, 100)$ as well.

- The players are required to choose their proposals simultaneously and both $(100, 100, 100)$ and $(150, 150, 0)$ are Nash equilibria for the players since unilateral deviation by any player will make all their utilities equal to zero.
- If player 3 has any influence, then player 3, being rational, would clearly try to influence players 1 and 2 to go for the equilibrium $(100, 100, 100)$.

However, there is one central assumption based on which the outcome $(100, 100, 100)$ loses credibility and cannot be justified. This is the *effective negotiation* assumption, which is a natural assumption to make, as articulated by Myerson in his book [1]. □

Myerson [1] defines effective negotiation as follows and asserts that effective negotiation is the key assumption that distinguishes cooperative game theory from non-cooperative game theory.

Definition 27.1 (Effective Negotiation). *The members of a coalition of players are said to* negotiate effectively *and are said to form an* effective coalition *if the players, on realizing that there is a feasible change in their strategies that would benefit them all, would all agree to actually make such a change unless such a change contradicts some agreements that some members of the coalition might have made with other players outside this coalition, in the context of some other equally effective coalition.*

The n-person Nash bargaining solution would be relevant if the only coalition that can negotiate effectively is the *grand coalition* that includes the whole of N. If other coalitions also can negotiate effectively, then the Nash bargaining solution is no longer relevant. This is because it ignores all information about the power of multi-player coalitions other than the grand coalition N. In Version 1 of the divide-the-dollar game, no coalition that is smaller than $\{1, 2, 3\}$ can guarantee more than 0 to its members. In Version 2, the coalition $\{1, 2\}$ could guarantee its members any payoff allocation that they could get in the coalition $\{1, 2, 3\}$.

Example 27.3 (Divide the Dollar Game - Version 3). This is a variant of Version 2 with the difference that players get a zero payoff unless player 1 and player 2 propose the same allocation or player 1 and player 3 propose the same allocation, in which case the players would get what is proposed. That is, for $i = 1, 2, 3$,

$$\begin{aligned} u_i(s_1, s_2, s_3) &= x_i \text{ if } s_1 = s_2 = (x_1, x_2, x_3) \text{ or } s_1 = s_3 = (x_1, x_2, x_3) \\ &= 0 \text{ otherwise} \end{aligned}$$

The bargaining problem (F, v) as in Versions 1 and 2 would describe the situation here and hence the Nash bargaining solution for this problem also is $x = (100, 100, 100)$. Much like in the case of Version 2, this solution also is unsatisfactory since players 1 and 2 together or players 1 and 3 together determine the payoff allocation. Player 1 is necessarily involved in both the above situations. So, we would expect the players to divide the payoff in a way that players 2 and 3 get the same payoff but this payoff should be less than the payoff that player 1 would get (since player 1 has to necessarily agree for a non-zero allocation). This leads to uncountably infinite number of possibilities, such as (120, 90, 90), (150, 75, 75), (200, 50, 50), (280, 10, 10), etc. One can even suggest an allocation (300, 0, 0) on the ground that player 1 is indispensable for a non-zero allocation. □

Example 27.4 (Divide the Dollar - Version 4 - Majority Voting Game). In this version, the players get a zero payoff unless there is some pair of players $\{1, 2\}, \{2, 3\}$, or $\{1, 3\}$ who propose the same allocation, in which case they get this allocation. That is,

$$\begin{aligned} u_i(s_1, s_2, s_3) &= x_i \text{ if } s_j = s_k = (x_1, x_2, x_3) \text{ for some } j \neq k \\ &= 0 \text{ otherwise} \end{aligned}$$

Here again, the Nash bargaining solution is $(100, 100, 100)$. This solution is clearly justified for this version because of symmetry and equal bargaining power of the players. Observe that this allocation is a Nash equilibrium. If we assume that every coalition can negotiate effectively, the analysis becomes quite interesting as seen below (citeMYERSON97).

- If players 1 and 2 negotiate effectively in the coalition $\{1, 2\}$, they can agree to the allocation $(150, 150, 0)$ which is attractive for both of them. Observe that this allocation is clearly a Nash equilibrium.
- If $(150, 150, 0)$ is the expected outcome, then player 3 would be eager to persuade player 1 or player 2 to form an effective coalition with him. For example, player 3 would be willing to negotiate an agreement with player 2 to both propose $(0, 225, 75)$ or, for example, $(0, 200, 100)$. This allocation is also a Nash equilibrium.
- If $(0, 225, 75)$ is the expected outcome in the absence of further negotiations, then player 1 would be willing to negotiate an agreement with player 3 to propose an allocation that is better for both of them, say, $(100, 0, 200)$. This allocation is again a Nash equilibrium.
- It turns out that in any equilibrium of this game, there is always at least one pair of players who would both do strictly better by jointly agreeing to change their strategies together and move to another equilibrium.

The above sequence of coalitional negotiations can go on endlessly. There are two possible ways in which the negotiations could conclude. The first possibility is the following. Let us say that a player, having negotiated an agreement as part of a coalition, cannot subsequently negotiate a different agreement with another coalition, that does not contain all the members of the first coalition. For example, if the grand coalition $\{1, 2, 3\}$ has negotiated the agreement $(100, 100, 100)$ before any two player coalition could negotiate separately, then no two player coalition can veto this outcome. Another example would be, if players 1 and 2 first negotiated an agreement $(150, 150, 0)$, then player 3 would be unable to increase this payoff by negotiating with player 1 or player 2 separately. It is clear that the order in which coalitions can negotiate will crucially determine the outcome of the game. The advantage lies with coalitions that get to negotiate earlier.

The second possibility is as follows. Suppose the negotiated agreements are only tentative and non-binding. Thus a player who negotiates in a sequential manner in various coalitions can nullify his earlier agreements and reach a different agreement with a coalition that negotiates later. Here the order in which negotiations are made and nullified will have a bearing on the final outcome. For example, let us say the order of negotiations is $\{1, 2\}$, $\{2, 3\}$, $\{1, 3\}$ and $\{1, 2, 3\}$. Here any agreement by $\{1, 2\}$ and $\{2, 3\}$ in that order to pay non-zero amount to player 2 can be overturned by the coalition $\{1, 3\}$ which might agree on $(150, 0, 150)$. Player 2 may not be able to make them agree to anything when the turn of coalition $\{1, 2, 3\}$ arises. As another example, assume that player 1 believes that, any negotiated agreement with player 2, would be overturned by player 3. Player 1 may first suggest $(100, 100, 100)$ and stick to it and refuse to agree for $(150, 150, 0)$. This he would do to prevent any possibility of his getting zero payoff. It is clear that coalitions that get to negotiate later hold the advantage in this scheme. □

In real-world scenarios where different coalitions could form, the number of ways in which negotiations could be conducted is mind-boggling and a systematic analysis of all scenarios may not be feasible. There are as many as $2^n - 1$ such coalitions possible and therefore there is a need for a theory of cooperative games that can

provide us with a clear sense of what to expect as a result of the balance of power among various coalitions. Such an order-independent theory will be extremely useful but will pose challenges in interpretation because ordering is often natural and important [1].

27.2 Transferable Utility Games (TU Games)

The assumption of *transferable utility* (TU) makes cooperative games easier to analyze and makes the analysis more tractable. The TU assumption implies that there is a commodity called money that the players can freely transfer among themselves such that the payoff of any player increases by one unit for every unit of money that he gets. With the assumption of transferable utility in place, the cooperative possibilities of a game can be described by a *characteristic function* $v : 2^N \to \mathbb{R}$, that assigns a number $v(C)$ to every coalition $C \subseteq N$. $v(\emptyset)$ is always taken to be zero. $v(C)$ is called the *worth* or *value* of the coalition C and it captures the total amount of transferable utility that the members of C could earn without any help from the players outside of C.

Definition 27.2 (TU Game). *A cooperative game with transferable utility is defined as the pair (N, v) where $N = \{1, \ldots, n\}$ is a set of players and $v : 2^N \to \mathbb{R}$ is a characteristic function, with $v(\emptyset) = 0$. We call such a game also as a game in coalition form, or game in characteristic form, or coalitional game, or TU game.*

Note. Under the assumption of transferable utility, specifying a single number for each coalition is enough to describe what allocations of utility can be achieved by the members of the coalition.

Non-Transferable Utility (NTU) Games

Games without transferable utility (also called NTU coalitional games or games in NTU coalitional form) are defined as follows.

Definition 27.3 (NTU Game). *An NTU coalitional game is a pair (N, V) where N is a set of players N and $V(\cdot)$ is a mapping with domain 2^N such that, for any coalition $C \subset N$,*

- *$V(C)$ is a non-empty closed and convex subset of $\mathbb{R}^{|C|}$, and*
- *The set $\{x : x \in V(C)$ and $x_i \geq v_i \; \forall i \in C\}$ is a bounded subset of $\mathbb{R}^{|C|}$, where*

$$v_i = \max\{y_i : y \in V(\{i\})\} < \infty \;\; \forall i \in N.$$

In the above definition, $V(C)$ is the set of expected payoff allocations that the members of coalition C could guarantee for themselves if they act cooperatively. An NTU game is a generalization of a TU game. In the remainder of the discussion, we will consider only TU games.

Examples of TU Games

Example 27.5 (Divide the Dollar Games). The Divide the Dollar - Version 1 game discussed in Example 27.1 has the following characteristic function.

$$\begin{aligned} v(\{1,2,3\}) &= 300 \\ v(\{1,2\}) = v(\{1,3\}) &= v(\{2,3\}) = 0 \\ v(\{1\}) = v(\{2\}) &= v(\{3\}) = 0 \end{aligned}$$

Version 2 of the game (Example 27.2) has the characteristic function:

$$\begin{aligned} v(\{1,2,3\}) = v(\{1,2\}) &= 300 \\ v(\{2,3\}) = v(\{1,3\}) &= 0 \\ v(\{1\}) = v(\{2\}) &= v(\{3\}) = 0 \end{aligned}$$

Version 3 of the game (Example 27.3) has the characteristic function:

$$\begin{aligned} v(\{1,2,3\}) = v(\{1,2\}) = v(\{1,3\}) &= 300 \\ v(\{1\}) = v(\{2\}) = v(\{3\}) = v(\{2,3\}) &= 0 \end{aligned}$$

Version 4 of the game (majority voting game) (Example 27.4) has the characteristic form:

$$\begin{aligned} v(\{1,2,3\}) = v(\{1,2\}) = v(\{2,3\}) = v(\{1,3\}) &= 300 \\ v(\{1\}) = v(\{2\}) = v(\{3\}) &= 0 \end{aligned}$$

□

Note. In the sequel, we will follow a slight abuse of notation while referring to the argument of the $v(\cdot)$ function: we shall drop the braces and commas while representing sets; for example, instead of $v(\{2,3\})$, we will simply write $v(23)$, etc., whenever there is no confusion.

Example 27.6 (A Voting Game). This example is taken from [2]. Consider that the Parliament of a certain Nation has four political parties 1, 2, 3, 4 with 45, 25, 15, 12 members respectively. To pass any bill, at least 51 votes are required. This situation could be modeled as a TU game with $N = \{1,2,3,4\}$ and

$$v(1) = v(2) = v(3) = v(4) = 0$$

$$v(12) = v(13) = v(14) = v(123) = v(124) = v(134) = v(234) = v(1234) = 1$$

$$v(23) = v(24) = v(34) = 0$$

□

Example 27.7 (Minimum Spanning Tree Game). Suppose multiple users are to be connected to a shared resource (for example, a power plant, a synchrotron, a radio telescope, a computational cluster, a logistics hub, a centralized drainage system, etc.). In order to utilize this resource, a user should either be directly connected to the facility or be connected to some other connected user. Figure 27.1(a) provides a picture of a typical

network of customers consisting of three players $1, 2, 3$ and a central shared facility F. We provide several typical examples of such a network below.

- Players $1, 2, 3$ could correspond to three different industrial units and the facility F could be a power plant from which the units draw their power. The edges in the graph correspond to high capacity power lines and the weights on the edges correspond to the cost of power distribution.
- Players $1, 2, 3$ could correspond to three different city corporations and F could correspond to a drainage system. The edges represent the drainage connections between the pairs of locations and the edge weights correspond to the cost of maintaining and using the connections.
- Players $1, 2, 3$ could correspond to pick up points in a logistics network and F could be a logistics hub. The edges correspond to the roads in the network and the edge weights correspond to the distances.

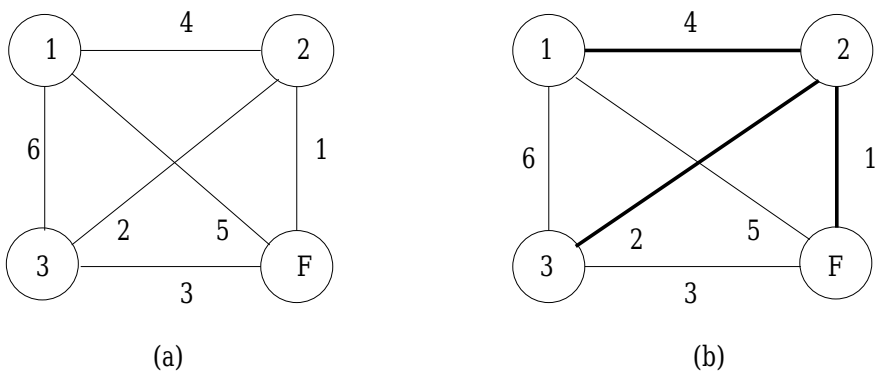

Fig. 27.1: A network of users connected to a central facility F

Suppose each player gets a benefit of 10 units when he gets connected to F directly or indirectly. Note that, given a coalition $C \subseteq \{1, 2, 3\}$, the minimum cost incurred by the members of coalition C to get connected to F is given by the sum of edge weights in a minimal spanning tree involving F and the members of C. For example, suppose $C = \{2, 3\}$. Then the minimum cost corresponding to C would be given by the sum of costs of edges in a minimal spanning tree involving the vertices $2, 3, F$. Such a minimal spanning tree consists of the edges $(2, F)$ and $(2, 3)$ and therefore has total cost 3. If $C = \{1, 2, 3\}$, then, the minimum cost would be the sum of edges of a minimal spanning tree involving the vertices $1, 2, 3, F$, which can be seen to be 7. Figure 27.1(b) depicts the minimal spanning tree involving $1, 2, 3, F$ through the thick edges.

We can in a natural way, define the worth of each coalition as the total benefit that accrues to the members of the coalition minus the minimum cost incurred in providing connectivity from all members of the coalition to F. For example, $v(23)$ would be $20 - 3$ since 20 is the total benefit that accrues to players 2 and 3 and the cost of the minimal spanning tree involving the nodes $2, 3, F$ is 3 units. We can now model the above situation as a TU game with characteristic function given by

$$v(\emptyset) = 0;\ v(1) = 10 - 5 = 5;\ v(2) = 10 - 1 = 9;\ v(3) = 10 - 3 = 7$$
$$v(12) = 20 - 5 = 15;\ v(13) = 20 - 8 = 12;\ v(23) = 20 - 3 = 17; v(123) = 30 - 7 = 23.$$

Such a game formulation would be useful in determining how the costs and profits could be shared by the players involved. □

Example 27.8 (A Logistics Game). Figure 27.2 shows a logistics network that provides connectivity between two important cities S and T. There are five logistics hubs A, B, C, D, E which are intermediate points from S to T. Transportation is provided by service providers 1, 2, 3, 4. Each edge in the network is labeled by two quantities namely the service provider and the cost of service. For example, the label $3, 15$ on the directed edge from A to B means that service provider 3 provides the logistics service from A to B at a cost of 15 units. Assume that movement from S to T fetches a revenue of 100 units. The objective is to choose an optimal path from S to T that maximizes the profit obtained while moving from S to T. We can formulate this as a cooperative game with $N = \{1, 2, 3, 4\}$ and with characteristic function

$$v(1) = v(2) = v(3) = v(4) = 0$$

$$v(12) = v(13) = v(14) = v(23) = v(24) = v(34) = v(234) = v(123) = 0$$

$$v(134) = 100 - 60 = 40$$

$$v(124) = 100 - 55 = 45$$

$$v(1234) = 100 - 35 = 65$$

A game such as above would be valuable in determining profit sharing protocols among the logistics providers. □

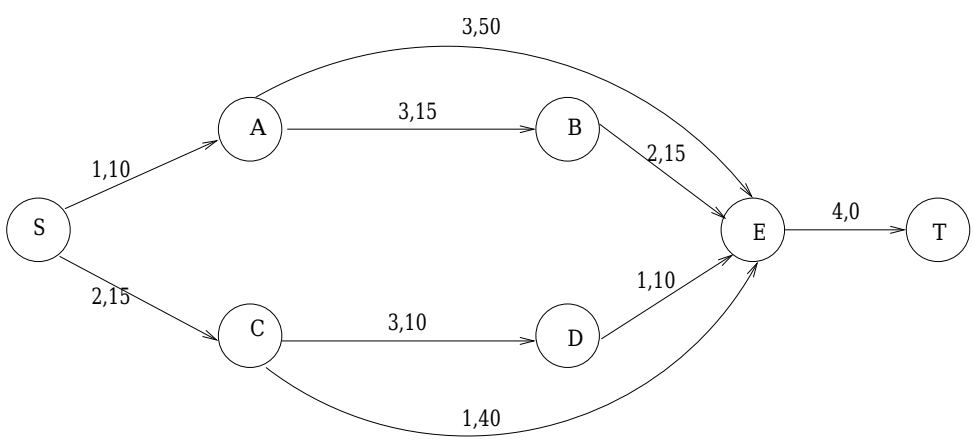

Fig. 27.2: A logistics network

Transforming Strategic Form Games to TU Games

If we have a strategic form game model of interactions among players and we wish to model cooperation among these players, there are several natural ways of defining

the characteristic function of a TU game. Three of the more common representations are: (1) Minimax representation, (2) Defensive equilibrium representation, and (3) Rational threats representation. For more details on these, we refer the reader to the book by Myerson [1] and the book by Maschler, Solan, and Zamir [3].

Imputations

Definition 27.4 (Imputation). *Given a TU game (N, v), an imputation is an allocation $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ that satisfies*

- *Individual Rationality :* $x_i \geq v(\{i\}) \quad \forall\, i \in N$
- *Collective Rationality :* $\sum_{i \in N} x_i = v(N)$

An imputation keeps all individual players happy and also distributes the total value of the grand coalition among the players (Pareto efficiency).

Definition 27.5 (Domination of Imputation). *An imputation $x = (x_1, \ldots, x_n)$ of a TU game (N, v) is said to dominate an imputation $y = (y_1, \ldots, y_n)$ if there exists a coalition C such that*

$$\sum_{i \in C} x_i \leq v(C); \text{ and } x_i > y_i \;\; \forall i \in C$$

Example 27.9 (Imputations and Domination). We make several observations concerning the notion of domination. Consider the majority voting game.

(1) The allocations $(150, 150, 0)$ and $(100, 0, 200)$ are both imputations while the allocations $(0, 0, 0)$, $(100, 50, 50)$ are not imputations. The imputation $(150, 150, 0)$ dominates the imputation $(100, 0, 200)$ since the coalition $\{1, 2\}$ is such that $150 > 100$ and $150 > 0$.
(2) Given two imputations x and y, it is possible that neither dominates the other. An immediate example would be the majority voting game where the imputation $x = (150, 150, 0)$ neither dominates nor is dominated by the imputation $y = (0, 150, 150)$.
(3) The relation of domination is not transitive and cycles of domination are possible. Again an immediate example would be majority voting game where imputation (0, 180, 120) dominates imputation (150, 150,0) which dominates imputation (90, 0, 210) which in turn dominates (0, 180, 120).

□

27.3 TU Games with Special Structure

In this section, we define several special classes of TU games.

Monotonic Games

A monotonic TU game is one in which as we add players to a coalition, the worth of the coalition does not decrease.

Definition 27.6 (Monotonic Game). *A TU game (N, v) is called monotonic if*

$$v(C) \leq v(D) \ \ \forall C \subseteq D \subseteq N.$$

Example 27.10. It can be verified that both the minimum spanning tree game (Example 27.7) and the logistics game (Example 27.8) are monotonic. In fact, all the examples we have discussed so far in this chapter are monotonic. A simple example of a non-monotonic game would be (N, v) with $N = \{1, 2\}$ and $v(1) = 10;\ v(2) = 20;\ v(12) = 15$. □

Superadditive Games

Superadditive games are TU games where the value of union of any two disjoint coalitions is at least as much as the sum of values of the individual coalitions, that is, two disjoint coalitions, on coming together, produce a non-negative additional value beyond the sum of the individual values.

Definition 27.7 (Superadditive Game). *A TU game (N, v) is said to be superadditive if*

$$v(C \cup D) \geq v(C) + v(D) \ \ \forall\, C, D \subseteq N \ \text{ such that } \ C \cap D = \phi$$

Example 27.11. The majority voting game with $N = \{1, 2, 3\}$ and v given by $v(1) = v(2) = v(3) = 0$ and $v(12) = v(13) = v(23) = v(123) = 300$ is a superadditive game. The minimum spanning tree game discussed in Example 27.7 is also a superadditive game. □

Example 27.12 (A Non-Superadditive Game). The following three player game is not superadditive.

$$v(1) = 10;\ v(2) = 15;\ v(3) = 20;\ v(12) = 20;\ v(13) = 30;\ v(23) = 35;\ v(123) = 40$$

□

Note. Note that a monotonic game need not be superadditive and a superadditive game need not be monotonic. It would be instructive to construct examples of games in all the four cases: (a) monotonic as well as superadditive (b) monotonic but not superadditive (c) non-monotonic but superadditive (d) neither monotonic nor superadditive.

Essential and Inessential Games

Definition 27.8 (Essential Superadditive Game). *A superadditive game (N, v) is said to be inessential if*

$$\sum_{i \in N} v(i) = v(N)$$

and essential otherwise.

If (N, v) is inessential then,

$$\sum_{i \in C} v(i) = v(C) \quad \forall\, C \subseteq N$$

Therefore $(v(1), v(2), \ldots, v(n))$ is the only possible imputation for an inessential game. On the other hand, there are infinitely many imputations possible for an essential game.

Strategic Equivalence of TU Games

Definition 27.9. Two TU games (N, v) and (N, w) are said to be strategically equivalent if there exist constants $c_1, c_2, \ldots, c_n$ and $b > 0$ such that

$$w(C) = b(v(C) + \sum_{i \in C} c_i) \quad \forall\, C \subseteq N$$

Intuitively, strategic equivalence means that the "dynamics" among the players would be identical in the two games.

An important result concerning strategic equivalence is that any superadditive, essential n-person characteristic form game G is strategically equivalent to a unique game with

$$N = \{1, 2, \ldots, n\}$$

$$v(1) = v(2) = \cdots = v(n) = 0; \;\; v(N) = 1$$

$$0 \leq v(C) \leq 1 \quad \forall\, C \subseteq N$$

This unique game is called the *0-1 normalization* of the original game.

Triangular Representation for Three Person Superadditive Games

The imputations in any three person game with $v(1) = v(2) = v(3) = 0$ and $v(N) = 1$ can be represented using an interesting triangular representation. This representation uses the following property: Suppose P is any point in an equilateral triangle with height h. Then the sum of the perpendicular distances from P to the three sides of the triangle is equal to h (see Figure 27.3).

Example 27.13. If we consider the majority voting game with $v(1) = v(2) = v(3) = 0; v(12) = v(13) = v(23) = v(123) = 1$ then any point $\rho = (x_1, x_2, x_3)$ inside the triangle represents an imputation because $x_1 \geq 0;\; x_1 \geq 0;\; x_2 \geq 0;\; x_3 \geq 0;\; x_1 + x_2 + x_3 = 1$. Figure 27.4 depicts several representative imputations. □

Example 27.14 (Bargaining in Majority Voting Game). Consider the majority voting game where we have $N = \{1, 2, 3\}$ and $v(1) = v(2) = v(3) = 0; v(12) = v(13) = v(23) = v(123) = 300$. Suppose we start with an allocation $(150, 150, 0)$ which indicates

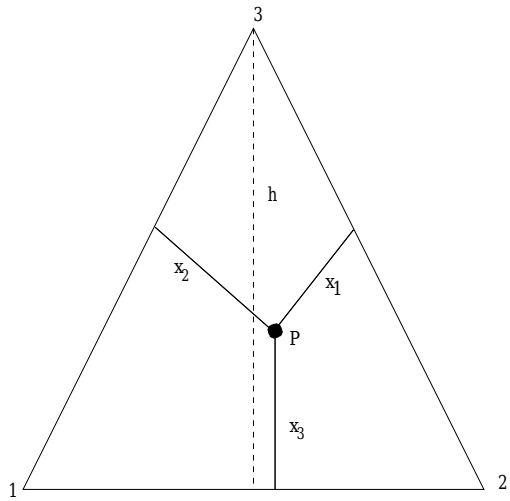

Fig. 27.3: An equilateral triangle, $x_1 + x_2 + x_3 = h$

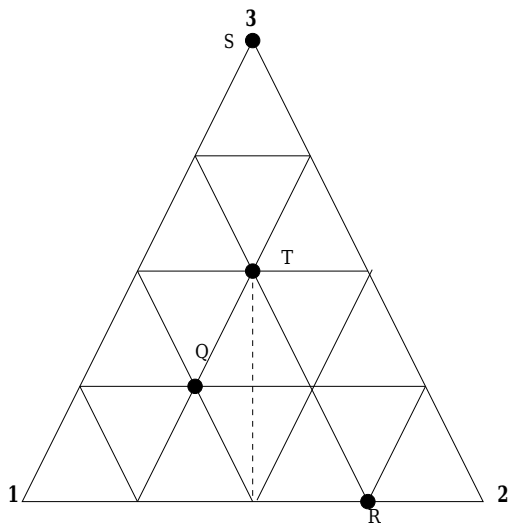

Fig. 27.4: $Q = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$; $R = (\frac{1}{4}, \frac{3}{4}, 0)$; $S = (0, 0, 1)$; $T = (\frac{1}{4}, \frac{1}{4}, \frac{1}{2})$ are imputations

a coalition between players 1 and 2. Player 3 can entice player 1 by proposing an allocation such as (180, 0, 120). Player 2 can now propose an allocation such as (0,120,180) and draw player 3 out of the coalition with player 1. In this particular game, bargaining can go on endlessly without any allocation being agreed upon. A graphical representation of this endless negotiation is presented in Figure 27.5. □

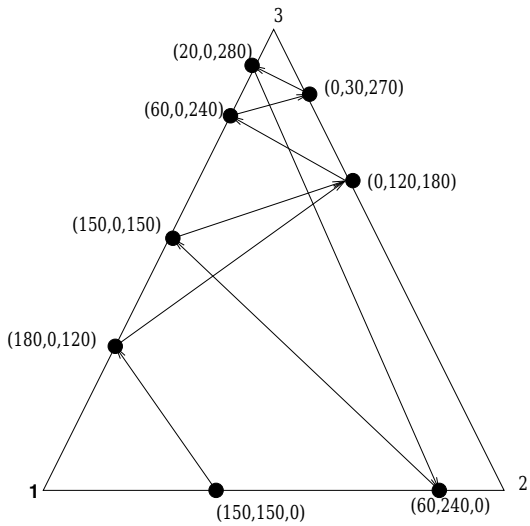

Fig. 27.5: Chain of unending negotiations in majority voting game

Convex Games

Convex games are those TU games where the marginal contribution of any player to a coalition rises as more players join the coalition. That is, players have an incentive to join larger coalitions than smaller coalitions. Formally convex games are defined as follows.

Definition 27.10 (Convex Game). *A TU game is said to be convex if*

$$v(C \cup D) + v(C \cap D) \geq v(C) + v(D) \;\; \forall C, D \subseteq N.$$

Note. In the above definition, suppose C and D are disjoint. Then we obtain

$$v(C \cup D) \geq v(C) + v(D) \;\; \forall\, C, D \subseteq N \text{ with } C \cap D = \emptyset$$

The above is precisely the definition of superadditivity. Thus every convex game is superadditive. The converse need not be true.

In the following propositions, we provide two equivalent definitions for convex games. The proof of these propositions is left as an exercise.

Proposition 27.1. *A TU game (N, v) is convex iff*

$$v(C \cup \{i\}) - v(C) \leq v(D \cup \{i\}) - v(D) \;\; \forall C \subseteq D \subseteq N \;\; \forall i \in N \setminus D.$$

Proposition 27.2. *A TU game (N, v) is convex iff*

$$v(C \cup E) - v(C) \leq v(D \cup E) - v(D) \;\; \forall C \subseteq D \subseteq N \;\; \forall E \subseteq N \setminus D.$$

27.4 Cost Games

The TU games we have discussed so far are referred to as *profit games* since the worth $v(C)$ of a coalition C represents the total (maximum) value that could be earned together by the members of the coalition without any help from other players (not belonging to the coalition C). If on the other hand, the worth of a coalition C represents the minimum total price the members of the coalition have to pay if the coalition were to form, then the resulting TU game is called a *cost game*. Cost games are complementary to profit games as illustrated in the following example. Often, cost games may be more natural to the given setting than profit games as shown in the following example.

Example 27.15 (Cost Game). Consider the minimum spanning tree game we discussed in Example 27.7. Let us define the worth of each coalition as the minimum cost incurred by the members of coalition C to get connected to F. Recall that, given a coalition $C \subseteq \{1, 2, 3\}$, the minimum cost incurred by the members of coalition C to get connected to F is given by the sum of edge weights in a minimal spanning tree involving F and the members of C. The characteristic function of the cost game in this situation is:

$$v(\emptyset) = 0;\ v(1) = 5;\ v(2) = 1;\ v(3) = 3$$

$$v(12) = 5;\ v(13) = 8;\ v(23) = 3; v(123) = 7.$$

Such a game formulation would be useful in determining how the costs could be shared by the players involved. □

27.5 Summary and References

The following are the salient aspects of this chapter.

- A natural extension of the Nash bargaining solution will not work for cooperative games containing three or more players since Nash bargaining only takes into account negotiations among all players in the grand coalition. This is illustrated clearly by the different versions of the divide the dollar problem.
- Coalition formation and negotiation among players in a cooperative game could involve intricate analysis since the negotiations might proceed endlessly and players may always find it in their benefit to leave a coalition and join a new coalition. Cooperative game theory provides a set of solution concepts to suggest outcomes that represent an acceptable and scientific end result of these complex negotiations.
- A transferable utility (TU) game or characteristic form game represents a cooperative game through a characteristic function $v : 2^N \to \mathbb{R}$ that associates a value or worth to each coalition of players. The worth of a coalition is the total utility the players of the coalition will be able to earn by themselves. Once a characteristic function is defined, the rest of the analysis can follow.

- A key issue in TU games is to derive the characteristic function $v(\cdot)$ to faithfully reflect the effect of cooperation among different subsets of players. Often, the physics of the underlying problem will naturally suggest a way of computing the characteristic function for different coalitions.
- An imputation in a TU game (N, v) is an allocation of payoffs to the players such that the sum of all payoffs is equal to $v(N)$ and the payoff for each player $i \in N$ is at least $v(\{i\})$.
- Superadditive games are TU games where the value of union of any pair of disjoint coalitions is no less than the sum of the values of the individual coalitions. Three person superadditive games can often be represented using a convenient triangular representation.
- Convex games are those in which a player's marginal contribution to any coalition is less than or equal to his marginal contribution to any superset of the coalition. Intuitively, marginal contribution of a player monotonically increases when more players join that coalition. Convex games are superadditive but not vice-versa.
- Cost games are TU games in which the the worth of a coalition is measured by the minimum cost incurred by the players rather than the maximum profit they can make by themselves. The TU games that are usually studied are called profit games and cost games are complementary to them.

The material in this chapter is mostly based on the books by Myerson [1] and Straffin [4]. The recent book by Maschler, Solan, and Zamir [3] is another valuable reference. The next three chapters deal with different solution concepts for TU games.

References

[1] Roger B. Myerson. *Game Theory: Analysis of Conflict.* Harvard University Press, Cambridge, Massachusetts, USA, 1997.

[2] Yoam Shoham and Kevin Leyton-Brown. *Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations.* Cambridge University Press, New York, USA, 2009, 2009.

[3] Michael Maschler, Eilon Solan, and Shmuel Zamir. *Game Theory.* Cambridge University Press, 2013.

[4] Philip D. Straffin Jr. *Game Theory and Strategy.* The Mathematical Association of America, 1993.

27.6 Exercises

(1) Provide an example of a a TU game in which every imputation is such that it is dominated by some other imputation.

(2) Construct examples of TU games for the following situations:

- Monotonic and superadditive

- Monotonic but not superadditive
- Superadditive but not monotonic
- Neither superadditive nor monotonic

(3) Show that the following game (called communication satellite game) is superadditive. Is it convex? (Source: [3]).

$$v(1) = 3;\ v(2) = 2;\ v(3) = 1;\ v(12) = 8;\ v(13) = 6.5;\ v(23) = 8.2;\ v(123) = 11.2$$

(4) Consider a TU game (N, v) where $N = \{1, 2, 3\}$ and

$$v(1) = v(2) = 1;\ v(3) = 2;\ v(12) = v(23) = v(13) = 4;\ v(123) = 5.$$

Is this game superadditive? Is it convex? (Source: [3]).

(5) Give an example of a TU game that is superadditive but not convex.

(6) Show that the 0-1 normalization of any three person constant sum game (that is, a TU game where, for every coalition $C \subseteq N$, $v(C) + v(N \setminus C)$ is a constant) is the three person majority voting game. (Source: [4]).

(7) Given two TU games (N, v) and (N, w), and a number $\lambda \in [0, 1]$, define a convex combination of the games as the game (N, z) where

$$z(C) = \lambda v(C) + (1 - \lambda)w(C)\ \forall C \subseteq N.$$

If (N, v) and (N, w) are convex, then would (N, z) also be convex? Prove your result.

(8) Prove Proposition 27.1: A TU game (N, v) is convex if and only if

$$v(C \cup \{i\}) - v(C) \leq v(D \cup \{i\}) - v(D)\ \forall C \subseteq D \subseteq N;\ \ \forall i \in N \setminus D.$$

(9) Prove Proposition 27.2: A TU game (N, v) is convex if and only if

$$v(C \cup E) - v(C) \leq v(D \cup E) - v(D)\ \ \forall C \subseteq D \subseteq N\ \ \forall E \subseteq N \setminus D.$$